

# 12.116 Matgeo

AI25BTECH11012 - Garige Unnathi

# Question

Let  $\mathbf{P}$  and  $\mathbf{Q}$  be two square matrices of same size. Consider the following statements

- ①  $\mathbf{PQ} = \mathbf{0}$  then  $\mathbf{P} = \mathbf{0}$  or  $\mathbf{Q} = \mathbf{0}$  or both
- ②  $\mathbf{PQ} = \mathbf{I}$  then  $\mathbf{QP} = \mathbf{I}$
- ③  $(\mathbf{P} + \mathbf{Q})^2 = \mathbf{P}^2 + 2\mathbf{PQ} + \mathbf{Q}^2$
- ④  $(\mathbf{P} - \mathbf{Q})^2 = \mathbf{P}^2 - 2\mathbf{PQ} + \mathbf{Q}^2$

Determine which statements are True and which are False

# Option 1

This option is False

This can be proven by a simple counter example :

$$\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \mathbf{Q} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad (1)$$

$$\mathbf{PQ} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (2)$$

## Option 2

Given  $\mathbf{PQ} = \mathbf{I}$

From this we can say that  $|\mathbf{P}|$  and  $|\mathbf{Q}|$  is not zero

That is inverse exists

$$\mathbf{Q} = \mathbf{P}^{-1} \quad (3)$$

$$\mathbf{QP} = \mathbf{P}^{-1}\mathbf{P} = \mathbf{I} \quad (4)$$

Hence this option is true

## Option 3

$$(\mathbf{P} + \mathbf{Q})^2 = (\mathbf{P} + \mathbf{Q})(\mathbf{P} + \mathbf{Q}) \quad (5)$$

$$= \mathbf{P}^2 + \mathbf{PQ} + \mathbf{QP} + \mathbf{Q}^2 \quad (6)$$

This equation will not be equal to  $\mathbf{P}^2 + 2\mathbf{PQ} + \mathbf{Q}^2$   
this is because Matrix multiplication is not commutative

$$\mathbf{PQ} \neq \mathbf{QP} \quad (7)$$

this option is false

# Option 4

Similarly we can also prove that Option 4 is false  
Hence only option 2 is true